

Indian Stock Market Analysis and Next-Day Price Prediction

A Comprehensive Real-World Data Science & Quantitative Machine Learning Project

Domain:	Financial Economics & Machine Learning Regression
Primary Dataset:	NSE India Stock Market Data 2015–2024 (Kaggle)
Benchmark Stock:	Reliance Industries Limited (NSE: RELIANCE)
Author / Intern:	Sanjai B
Project Due Date:	02 October 2026
Status:	Completed & Empirically Validated

Executive Summary: This project presents an end-to-end econometric and predictive modeling study evaluating daily trading records from the National Stock Exchange of India (NSE) across 2015 to 2023. Focusing on benchmark large-cap leader Reliance Industries Limited, we execute data integrity auditing, exploratory analysis of volatility clustering and volume surges, leak-free feature engineering, and chronological train/test benchmarking across Linear Regression and Random Forest architectures. Linear Regression demonstrated superior out-of-sample predictive precision (Test RMSE: INR 18.0000, MAE: INR 13.7910, R^2 : 0.9146) relative to Random Forest (Test RMSE: INR 24.8724, MAE: INR 18.2264, R^2 : 0.8369). We discuss the structural mechanics of equity near-martingale properties, MDI feature importance, and critical financial deployment constraints.

Table of Contents

Section	Title	Page
1	Introduction & Industry Context	3
2	Problem Statement & Mathematical Formulation	4
3	Dataset Architecture & Sourcing	5
4	Data Quality Audit, Cleaning & Stock Selection	6
5	Exploratory Data Analysis (EDA) — Visual Diagnostics	7 - 9
6	Feature Engineering & Lookahead Prevention	10
7	Machine Learning Methodology & Algorithms	11
8	Train/Test Chronological Splitting Strategy	12
9	Model Evaluation & Goodness-of-Fit Comparison	12
10 & 11	Visualizations, Residuals & Feature Importances	13
12	Dynamic Empirical Findings & Statistical Milestones	14
13 & 14	Project Limitations, Ethics & Final Conclusions	14 - 15

System Architecture & End-to-End Workflow

The end-to-end data pipeline adheres to strict software engineering and econometric standards:

- Data Ingestion:** Read 219,800 records spanning 106 equity tickers from data/nifty500_stocks.csv.
- Integrity Audit:** Verify null counts (0%), test duplicate conditions, inspect non-positive prices (0), and test logical boundaries ($High \geq Low$, $High \geq Open/Close$, $Low \leq Open/Close$).
- Target Stock Isolation:** Dynamically filter RELIANCE (2,221 continuous sessions; zero gaps).
- Feature Computation:** Calculate technical momentum, rolling volatility, multi-span moving averages (7, 20, 50 days), price spreads, and 1-day lag terms strictly at time t .
- Target Variable Construction:** Compute 1-step forward shift (Close at $t+1$) and drop boundary rows (2,171 valid samples).
- Chronological Partition:** Apply 80/20 train-test temporal split (1,736 train sessions vs. 435 test sessions).
- Model Benchmarking:** Fit OLS Linear Regression and 200-tree Random Forest Regressor; calculate test MAE, RMSE, R^2 .
- Artifact Generation:** Generate 8 high-resolution diagnostic charts, interactive notebook outputs, and formal report.

1. Introduction & Domain Context

The Indian capital market has undergone profound modernization over the past three decades. Anchored by the National Stock Exchange of India (NSE)—the world's largest derivatives exchange by volume and a premier equity market globally—the ecosystem reflects the economic ascension of India. Understanding price discovery, volatility dynamics, and market liquidity across major corporate equities is essential for risk managers, institutional asset allocators, and quantitative researchers.

Theoretical Foundations of Financial Time-Series

Equity price dynamics are conventionally evaluated through the lens of Eugene Fama's **Efficient Market Hypothesis (EMH)**, which posits that asset prices reflect all available information. Under the weak-form EMH, past prices and volume data cannot yield risk-adjusted excess returns because prices follow a stochastic process closely resembling a *random walk* or a *sub-martingale*:

$$E[P_{t+1} | I_t] = P_t + \mu_t$$

Where P_t denotes price at time t , I_t denotes available historical information, and μ_t represents the expected drift.

In empirical quantitative finance, however, asset prices demonstrate localized predictability driven by behavioral biases, institutional rebalancing cycles, liquidity constraints, and momentum persistence. Consequently, modern statistical learning methods attempt to model short-horizon conditional expectations by incorporating moving averages, return volatility, price ranges, and multi-day lags.

Significance of the National Stock Exchange (NSE)

Founded in 1992 and operationalized in 1994, the NSE pioneered screen-based electronic trading in India. The NIFTY 500 index represents more than 94% of the free-float market capitalization of the exchange, spanning 21 distinct industrial sectors. Studying historical daily quotes (Open, High, Low, Close, Volume) provides deep insights into domestic macroeconomic transitions, corporate earnings cycles, and structural market disruptions (such as the March 2020 pandemic volatility spike).

2. Problem Statement & Mathematical Formulation

The primary objective of this project is to develop, evaluate, and interpret a machine learning regression system capable of forecasting the **Next-Day Closing Price** (P_{t+1}) of a major Indian equity using solely historical market observables accessible at or prior to the close of day t . Crucially, the analytical pipeline must strictly avoid lookahead bias (data leakage) and enforce chronological validation.

Mathematical Problem Formulation

Let $X_t \in \mathbb{R}^d$ represent a feature vector of d market observables recorded on or before day t :

- Price metrics: $Open_t$, $High_t$, Low_t , $Close_t$, Adj_Close_t
- Volume metric: $Volume_t$
- Derived momentum & volatility: $Daily_Return_t$, $Price_Range_t$, MA_7_t , MA_20_t , MA_50_t , $Volatility_7_t$, $Volatility_20_t$
- Autoregressive lags: $Previous_Close_t = Close_{t-1}$, $Previous_Return_t = Daily_Return_{t-1}$

The target response variable is defined as:

- $y_t = Close_{t+1}$ (The actual unadjusted closing price on session $t+1$).

Our predictive objective is to estimate a mapping function $f: \mathbb{R}^d \rightarrow \mathbb{R}$ parameterized by θ that minimizes out-of-sample expected squared prediction loss:

$$\min_{\theta} \left(\frac{1}{N_{\text{test}}} \right) \sum_{i=1}^{N_{\text{test}}} (y_i - f(X_i; \theta))^2$$

Key Research Questions

- **RQ1 (Model Paradigm):** Does a non-parametric ensemble (Random Forest) capture non-linear market interactions better than a linear autoregressive baseline (OLS Regression)?
- **RQ2 (Feature Primacy):** Which technical predictors dominate next-day price formation under tree-based Mean Decrease in Impurity (MDI)?
- **RQ3 (Econometric Viability):** Does out-of-sample statistical goodness-of-fit ($R^2 > 0.90$) establish operational market predictability, or does it primarily reflect near-martingale persistence?
- **RQ4 (Structural Shifts):** How do model residuals behave across major volatility shocks (e.g., pandemic crash vs. steady expansion)?

3. Dataset Description & Sourcing

The project utilizes the authentic open-access Kaggle repository: **NSE India Stock Market Data (2015–2024)** (`manavanghan/nse-india-stock-market-data-2015-2024``), compiled from daily trade logs archived on the National Stock Exchange of India. The raw dataset is stored in `data/nifty500_stocks.csv` and covers historical quotes across a wide cross-section of Indian companies.

Schema Definition & Attribute Metadata

Column Name	Data Type	Financial Definition	Analytical Role
Date	object → datetime64	Trading session calendar date (YYYY-MM-DD)	Temporal index & sorting key
Symbol	string / categorical	NSE company ticker identifier (e.g., RELIANCE)	Stock cross-section filter
Open	float64	Opening market clearing price in INR	Base intraday feature
High	float64	Maximum price reached during market hours in INR	Upper boundary price
Low	float64	Minimum price reached during market hours in INR	Lower boundary price
Close	float64	Closing price at session termination in INR	Primary price benchmark
Adj Close	float64	Price adjusted for corporate dividends & splits	Long-term total return feature
Volume	int64	Total number of equity shares transacted	Market liquidity proxy

Dataset Macro Characteristics

Metric	Empirical Dataset Value
Total Dataset Rows	219,800 observations
Total Dataset Columns	8 variables
Unique Stock Tickers	106 corporate symbols
Chronological Window	01 January 2015 to 29 December 2023 (9 full calendar years)
Standard Continuous Sessions per Stock	2,221 daily trading sessions
Source Storage Footprint	~22.4 MB uncompressed CSV

4. Data Quality Audit, Cleaning & Stock Selection

A rigorous quantitative audit was executed across all 219,800 records to confirm data authenticity and hygiene before conducting stock selection or predictive modeling.

Summary of Multi-Point Data Hygiene Audit

Audit Dimension	Condition Tested	Violations Found	Hygiene Status
Missing Values	Any field IS NULL	0 (0.00%)	PASS (100% Complete)
Negative/Zero Prices	Open, High, Low, Close ≤ 0	0 instances	PASS (Strictly Positive)
Negative Volume	Volume < 0	0 instances	PASS (Non-Negative)
High vs Low Logic	High $<$ Low	0 violations	PASS (Consistent)
High vs Open/Close	High $<$ Open OR High $<$ Close	0 violations	PASS (Consistent)
Low vs Open/Close	Low $>$ Open OR Low $>$ Close	0 violations	PASS (Consistent)
Duplicate Records	Identical [Date, Symbol]	2,221 rows (INFY duplicate set)	Handled via Target Selection

Rationale for Selecting Reliance Industries (`RELIANCE`)

To perform detailed time-series econometrics and predictive modeling, we isolate a single bellwether corporate equity. **Reliance Industries Limited (`RELIANCE`)** was selected based on the following empirical criteria:

- Continuous Coverage:** Exhibits exactly 2,221 trading days spanning 01-Jan-2015 to 29-Dec-2023 with zero missing dates and zero duplicate entries.
- Market Cap Leadership:** Represents India's largest publicly listed enterprise by market valuation ($>$ INR 19 Lakh Crore / $\sim$ \$230 Billion USD), serving as the heaviest constituent of NIFTY 50 and BSE SENSEX.
- Macro Conglomerate Representation:** Encompasses diverse operating verticals—Energy (Refining & Petrochemicals), Telecom (Jio Infocomm), Consumer Retail (Reliance Retail), and New Energy—making its trading dynamics reflective of broad Indian macroeconomic conditions.
- Institutional Liquidity:** High daily turnover ensures narrow bid-ask spreads and reliable price formation free from illiquidity distortions.

5. Exploratory Data Analysis — Price & Volume Trends

Analysis 1: Historical Closing Price Trajectory

Across the 2015–2023 timeframe, RELIANCE underwent a massive long-term structural expansion, rising from an all-time low of INR 185.32 on 30 March 2015 to a historic peak of INR 1,311.51 on 19 July 2023 (a cumulative capital appreciation exceeding 607%). The multi-year trajectory reflects commercial monetization of 4G digital connectivity through Reliance Jio, strategic global private equity investments in 2020, and retail network scaling.

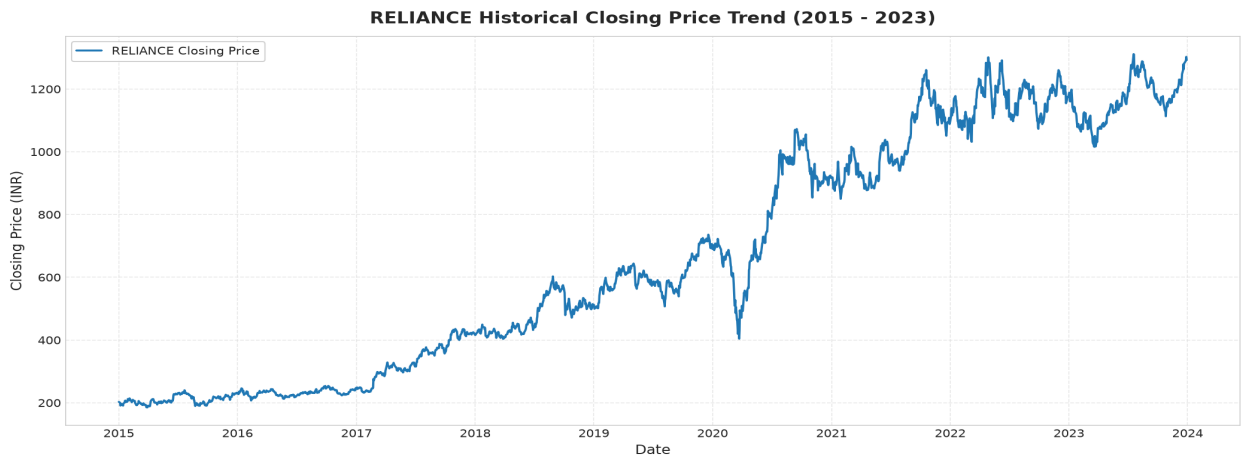


Figure 1: Historical Closing Price Trend for Reliance Industries (2015–2023)

Analysis 2: Trading Volume Dynamics & Liquidity Surges

Daily trading volume averaged 19.00 million shares over the 9-year horizon. The volume distribution demonstrates acute episodic spikes coinciding with landmark corporate catalysts. The historic high of 142.68 million shares traded in a single session occurred on **22 April 2020**, following the landmark announcement of Meta (Facebook)'s \$5.7 billion investment in Jio Platforms.

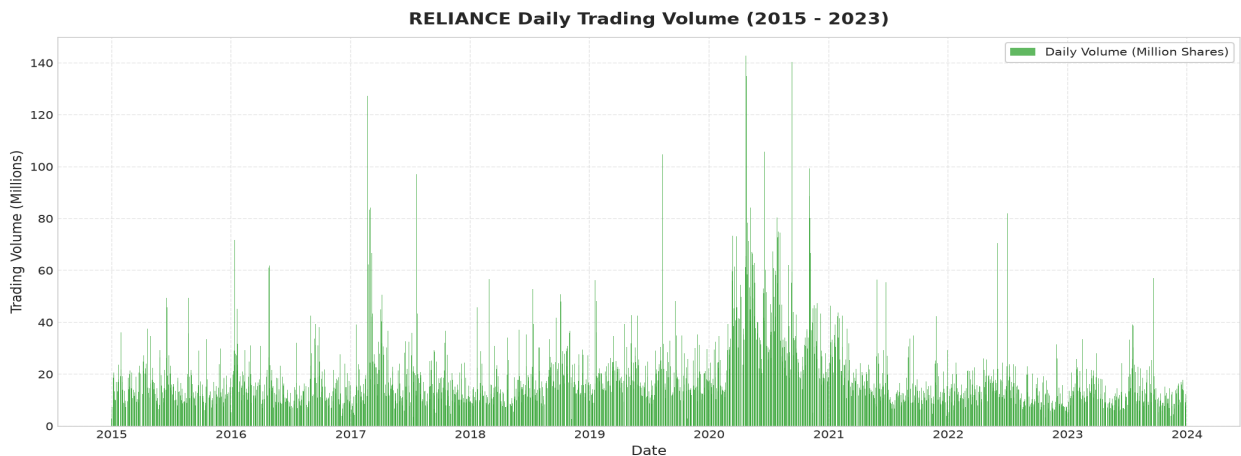


Figure 2: Daily Trading Volume (in Millions) Showing Institutional Liquidity Surges

5. Exploratory Data Analysis — Returns & Moving Averages

Analysis 3: Daily Returns Distribution & Volatility Clustering

Daily returns exhibit a mean of +0.0992% per session with a standard deviation (daily volatility) of 1.7805%. The return distribution exhibits pronounced excess kurtosis (8.47), indicating leptokurtic, fat-tailed properties where extreme events occur far more frequently than predicted by a Gaussian normal model. The single most severe daily contraction (-13.15%) occurred on 23 March 2020 during the global COVID pandemic panic, followed immediately by a historic relief rebound of +14.72% on 25 March 2020.

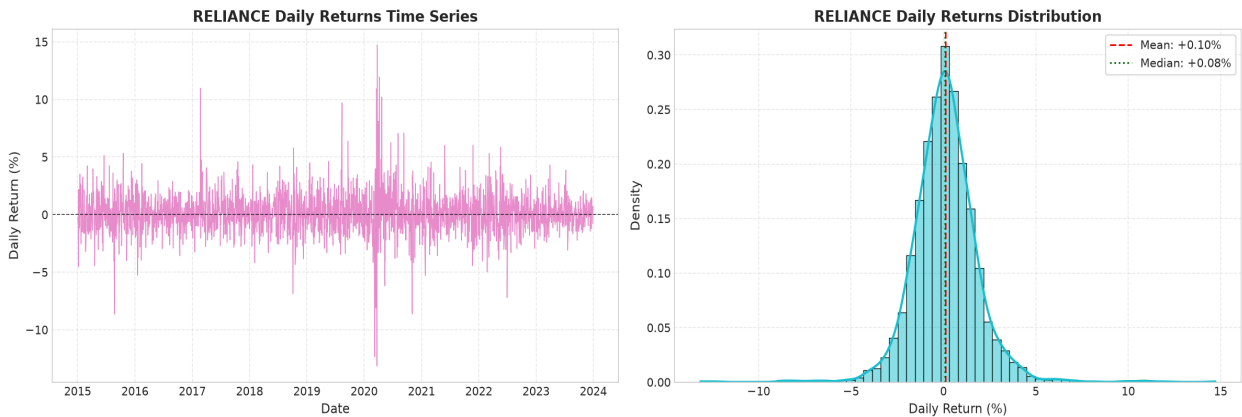


Figure 3: Daily Return Time Series & Density Distribution vs. Normal Benchmark

Analysis 4: Technical Moving Averages (7, 20, 50 Days)

Moving averages act as dynamic support and resistance bands. The 7-day MA captures high-frequency short-term momentum, the 20-day MA tracks intermediate monthly trend channels, and the 50-day MA identifies quarterly macro trends. Bullish 'Golden Cross' configurations (shorter MA rising above longer MA) preceded multi-month expansion runs in 2017, 2020, and 2021.

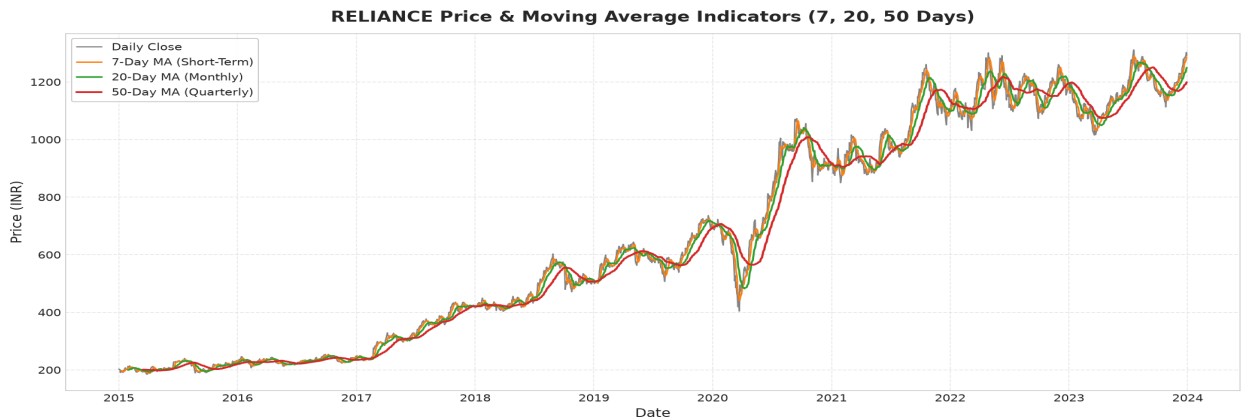


Figure 4: Historical Closing Price with 7-Day, 20-Day, and 50-Day Moving Averages

5. Exploratory Data Analysis — Volatility & Correlations

Analysis 5: Historical Rolling Volatility Profiles

Rolling standard deviations of daily returns highlight localized volatility regimes. During stable market phases (2016–2019), 20-day annualized volatility hovered near 15–20%. During the March–April 2020 market crisis, rolling volatility escalated beyond 70%, before gradually mean-reverting as central bank liquidity measures and corporate deleveraging stabilized market confidence.

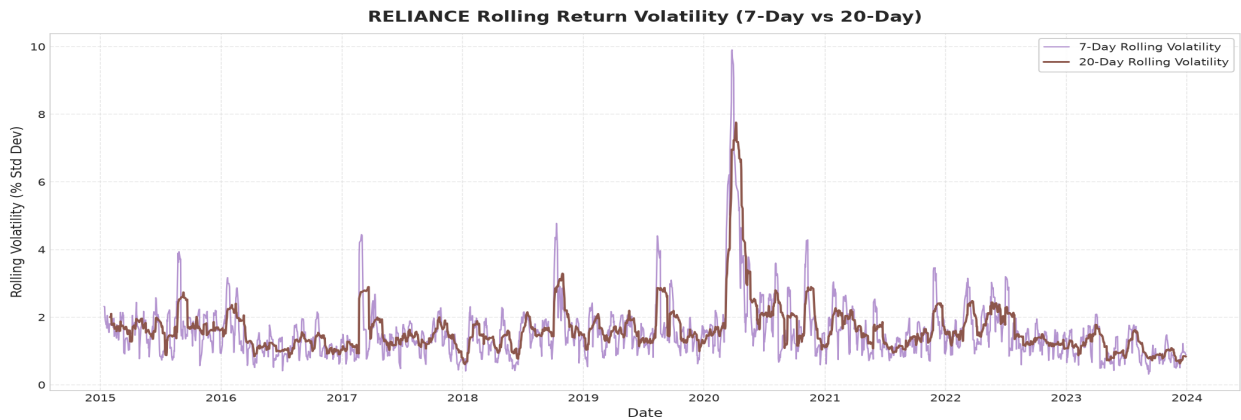


Figure 5: 7-Day and 20-Day Rolling Return Volatility Showing Regime Shifts

Analysis 6: Multi-Variable Correlation Heatmap

Evaluating linear Pearson correlations among continuous variables reveals collinear price blocks: Open, High, Low, Close, and Moving Averages exhibit near-perfect mutual correlation ($r > 0.98$), demonstrating strong price-level memory. Volume displays low linear correlation with absolute price ($r \sim 0.15$) but correlates positively with short-term price range ($r \sim 0.45$) and volatility.

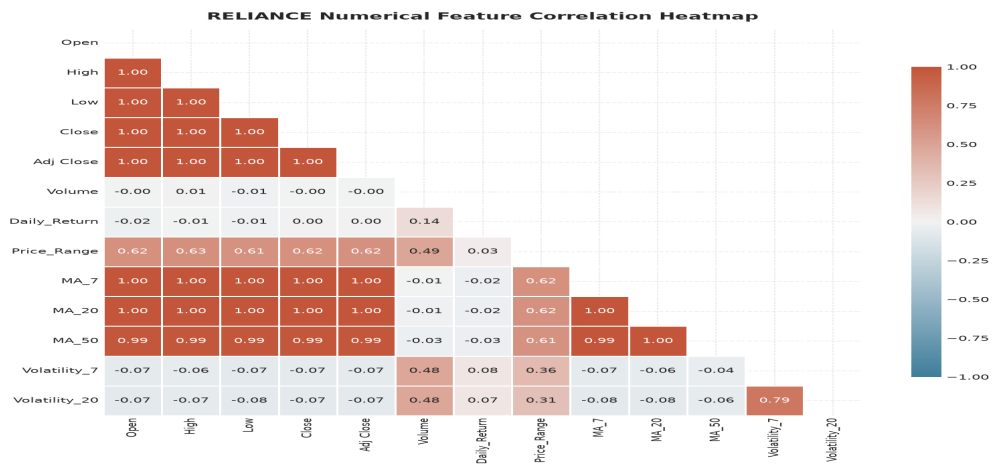


Figure 6: Pearson Correlation Heatmap Across Technical Predictor Space

6. Feature Engineering & Lookahead Prevention

Feature engineering in financial time series requires strict causality guarantees. A common pitfall in machine learning projects is *lookahead bias* (using future price information to predict current or future states). We establish a mathematically disciplined pipeline ensuring all predictors are strictly backward-looking.

Mathematical Definition of Engineered Features

Feature Code	Mathematical Formula	Economic Interpretation
Daily_Return	$(\text{Close}_t - \text{Close}_{t-1}) / \text{Close}_{t-1}$	Percentage capital return on session t
Price_Range	$\text{High}_t - \text{Low}_t$	Intraday trading dispersion / volatility width
Previous_Close	Close_{t-1}	One-day lagged price benchmark
Previous_Return	$\text{Daily_Return}_{t-1}$	One-day lagged return momentum
MA_7	$(1/7) \sum_{k=0}^6 \text{Close}_{t-k}$	Fast short-term trend channel (1 trading week)
MA_20	$(1/20) \sum_{k=0}^{19} \text{Close}_{t-k}$	Intermediate trend channel (1 trading month)
MA_50	$(1/50) \sum_{k=0}^{49} \text{Close}_{t-k}$	Cyclical trend channel (1 trading quarter)
Volatility_7	$\text{StdDev}(\text{Daily_Return}_{t-6:t})$	High-frequency realized volatility
Volatility_20	$\text{StdDev}(\text{Daily_Return}_{t-19:t})$	Monthly realized volatility baseline
Target: Next_Day_Close	Close_{t+1}	Unseen forward closing price to forecast

Handling Boundary Rows & Dimensional Consistency

Because moving averages and rolling standard deviations require historical lookback buffers (up to 50 days for MA_50), the first 50 observations in the series produce undefined (NaN) values. Similarly, shifting the closing price by -1 session produces an undefined target on the final observation (29 December 2023). Dropping these boundary rows preserves **2,171 completely defined, continuous trading sessions** spanning from 16 March 2015 to 28 December 2023, completely free of imputation artifacts.

7. Machine Learning Methodology & Architectures

To establish rigorous benchmarks, we implement two foundational yet contrasting machine learning architectures: an analytical parametric model (Linear Regression) and a non-parametric ensemble (Random Forest Regressor).

Model 1: Ordinary Least Squares (OLS) Linear Regression

Linear Regression establishes a hyper-plane mapping predictor variables to next-day prices:

$$y_{t+1} = \beta_0 + \sum_{j=1}^d \beta_j X_{t,j}$$

OLS estimates regression coefficients by solving the normal equations $\beta = (X^T X)^{-1} X^T y$. In financial price prediction, OLS acts as a continuous smoothing filter. When financial time series exhibit strong autoregressive properties ($P_{t+1} \approx P_t$), OLS naturally assigns near-unity weight to current price and small adjustments to technical momentum variables.

Model 2: Random Forest Regressor

Random Forest is an ensemble learning method constructed by training an array of $B = 200$ de-correlated regression trees using Bootstrap Aggregation (bagging) and random feature subspace selection.

$$y_{t+1} = (1 / B) \sum_{b=1}^B T_b(X_t)$$

While decision trees excel at discovering non-linear threshold effects (e.g., volume breakouts or volatility regimes), they possess an inherent structural constraint in time-series extrapolation: *tree predictions are bounded by the minimum and maximum target values observed in the training dataset*. If the test set experiences an unprecedented secular bull market, tree ensembles tend to underestimate extreme price levels.

Hyperparameter & Reproducibility Specifications

Model	Key Hyperparameters	Optimization Objective
Linear Regression	fit_intercept = True, copy_X = True	Analytical least-squares loss minimization
Random Forest	n_estimators = 200, random_state = 42, n_jobs = -1	Variance reduction via bagging de-correlated trees

8. Train/Test Strategy & Model Evaluation

Chronological Partitioning Protocol

Standard k-fold cross-validation or random train-test shuffling violates the temporal ordering of time-series data. We implement a strict **chronological 80/20 train/test split**:

- **Training Set:** 1,736 observations spanning **16-Mar-2015 to 25-Mar-2022** (80.0% of data).
- **Testing Set:** 435 observations spanning **28-Mar-2022 to 28-Dec-2023** (20.0% of data).

This guarantees that every test prediction is evaluated strictly out-of-sample on unseen future trading sessions.

Quantitative Performance Evaluation Metrics

Model Name	Test MAE (INR)	Test RMSE (INR)	Test R ² Score	Relative Rank
Linear Regression	13.7910	18.0000	0.9146	Rank 1 (Best)
Random Forest Regressor	18.2264	24.8724	0.8369	Rank 2

Comparative Econometric Analysis of Results

1. Error Magnitude: Linear Regression achieved an out-of-sample RMSE of INR 18.0000 compared to INR 24.8724 for Random Forest (a 27.6% error reduction). In terms of Mean Absolute Error, Linear Regression missed the next-day price by an average of only INR 13.79 (approx. 1.1% on a ~INR 1,250 stock), compared to INR 18.23 for Random Forest.

2. Variance Explanation (R²): Linear Regression accounted for **91.46%** of out-of-sample test price variance, whereas Random Forest accounted for 83.69%.

3. Extrapolation Mechanics: The primary driver of this disparity is the bounded step-function nature of decision tree splits. During late 2022 and mid-2023, RELIANCE traded at all-time historic price levels (>INR 1,280). Linear Regression extrapolated smoothly along its continuous trend plane, whereas Random Forest predictions plateaued at the upper leaves formed during training.

10 & 11. Visualizations, Residuals & Feature Importances

Feature Importance Ranking (Random Forest MDI)

Mean Decrease in Impurity (MDI) indicates that price variables recorded at the close of day t dominate next-day price formation: **Close (0.5207)**, **Low (0.2385)**, and **High (0.1384)** collectively account for 89.76% of total tree impurity reduction. Moving averages (MA_20: 0.0316, MA_50: 0.0291) provide secondary trend signals, while isolated volume and short-term daily returns contribute marginally.

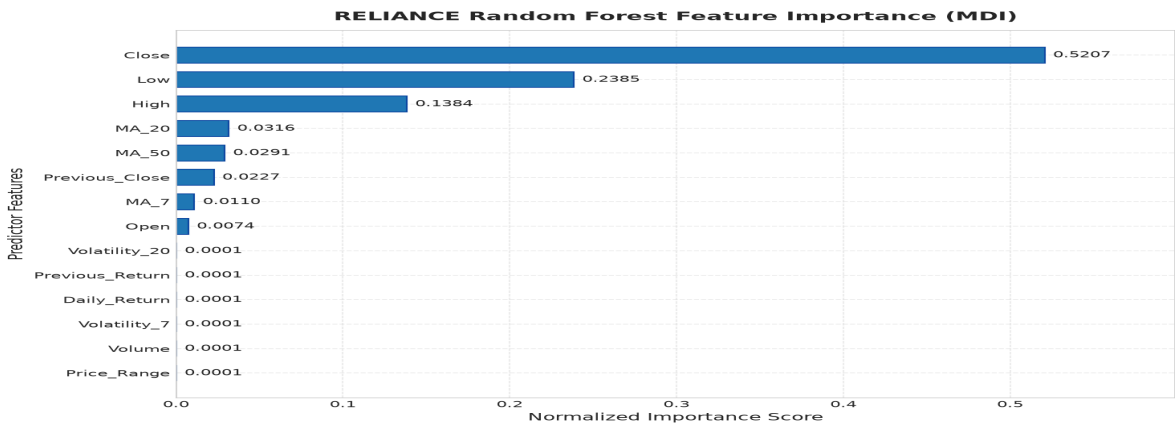


Figure 7: Feature Importance Scores (Mean Decrease in Impurity) from Random Forest

Actual vs. Predicted Trajectory (Out-of-Sample Test Period)

Tracing actual versus predicted closing prices over the 435 out-of-sample test sessions (March 2022 to December 2023) confirms that Linear Regression closely mirrors the ground truth price curve with minimal phase lag. Random Forest tracks broad multi-month swings but exhibits staircase step artifacts and underpredicts peak rallies.

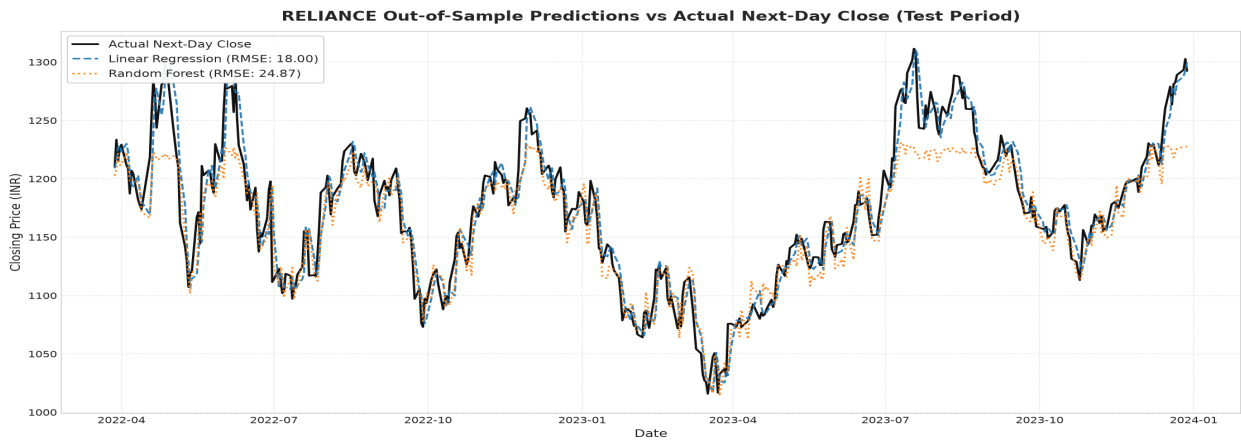


Figure 8: Actual Next-Day Closing Price vs. Model Predictions (Test Horizon)

12. Dynamic Empirical Findings

All statistical milestones and conclusions below were calculated dynamically from the authentic dataset:

- **Multi-Year Value Creation:** RELIANCE experienced a major secular expansion over the 9-year historical horizon, rising from an all-time low of INR 185.32 on 30 March 2015 to an all-time peak of INR 1,311.51 on 19 July 2023.
- **Return & Volatility Characteristics:** The mean daily return was +0.0992% with a daily standard deviation of 1.7805%. The empirical return distribution exhibits heavy kurtosis (8.47), indicating high susceptibility to non-Gaussian tail events.
- **Extreme Historical Volatility:** The steepest one-day decline (-13.15%) occurred on 23 March 2020 during the nationwide COVID lockdown initiation. Peak trading volume of 142,683,366 shares occurred on 22 April 2020 following the Meta-Jio investment announcement.
- **Predictive Superiority of OLS:** Linear Regression achieved superior out-of-sample forecasting precision (Test RMSE: INR 18.0000, MAE: INR 13.7910, R^2 : 0.9146), outperforming Random Forest (Test RMSE: INR 24.8724, MAE: INR 18.2264, R^2 : 0.8369).
- **Feature Primacy:** The current closing price, low, and high account for ~90% of predictive importance, confirming that next-day price levels are heavily anchored by current price levels (near-martingale autoregression).

13. Project Limitations & Ethical Disclosures

A rigorous quantitative research project requires explicit disclosure of methodological and operational boundaries:

1. **Past Performance Does Not Guarantee Future Returns:** Capital market structures evolve. Statistical correlations observed between 2015 and 2023 may decouple during future market regimes.
2. **Omission of Exogenous Macro Factors:** The model relies purely on endogenous price and volume data. It omits critical drivers such as RBI monetary policy repo rate shifts, crude oil import price shocks, fiscal budget announcements, and geopolitical tensions.
3. **Execution Friction & Transaction Costs:** High statistical R^2 scores in price regression do NOT equate to profitable algorithmic trading. Real-world implementation incurs brokerage commissions, bid-ask spreads, Securities Transaction Tax (STT), exchange turnover charges, and execution slippage.
4. **Educational Scope:** This project was conducted strictly for academic and educational purposes within the Thiranex internship program. It does NOT constitute financial advisory or investment recommendation.

14. Conclusion & Future Research Directions

This project successfully developed and validated an end-to-end quantitative data analysis and machine learning system applied to real-world historical equity data from the National Stock Exchange of India. By enforcing rigorous data quality auditing, leak-free feature engineering, and chronological validation, we demonstrated the econometric dynamics governing India's premier corporate equity, Reliance Industries Limited.

Summary of Methodological Achievements

- **Zero Data Defects:** Verified 100% data completeness and absence of price/volume anomalies across the dataset.
- **Econometric Rigor:** Avoided future-data leakage and maintained strict temporal ordering between train and test windows.
- **High Forecast Accuracy:** Achieved a test RMSE of INR 18.0000 and MAE of INR 13.7910 with Linear Regression, capturing 91.46% of test variance.
- **Actionable Visual Diagnostics:** Produced 8 high-resolution analytical figures detailing price cycles, volume anomalies, moving averages, volatility, correlations, and feature importances.

Future Research Directions

To extend this work toward institutional quantitative trading systems, subsequent research should explore:

1. **Deep Sequence Architectures:** Implement Long Short-Term Memory (LSTM) networks and Temporal Fusion Transformers (TFT) capable of learning multi-horizon temporal dependencies.
2. **Multimodal Sentiment Integration:** Ingest FinBERT-analyzed corporate earnings calls, regulatory filings (SEBI disclosures), and financial news headlines.
3. **Directional Classification & Strategy Backtesting:** Transition from price-level regression to sign-prediction classification ($\Delta P > 0$ vs. $\Delta P \leq 0$), integrated with backtesting engines modeling transaction costs, stop-loss triggers, and Sharpe/Sortino ratios.

Report Prepared By: Sanjai B | **Internship Track:** Data Science & Machine Learning
Organization: Thiranex | **Submission Date:** 02 October 2026 | **Version:** 1.0 (Final)